EXPERIMENT - 10(a)

28/9/26. INTERNETWORKING WITH ROUTERS.

AIM:

Internetworking with routers in CISCO Packet Tracer simulator

PROCEDURE:

Step 1 :- Configuring Router 1

1) Open CISCO Packet Tracer
2) Select the router and open the CLI. and press enter to access the command prompt
3) Enter privileged mode using enable
4) Enter global configuration mode using configure
5) Configure Fast Ethernet 0/0 with:
   - IP Address : 192.168.20.1.
   - Subnet mask : 255.255.255.0.
6) Use no shutdown to activate the Interface
7) Configure fastEthernet 0/1 with:
   - IP Address : 192.168.20.1
   - Subnet mask : 255.255.255.0
8) Use no shutdown to activate the Interface.

Step 2: Configuring PCs

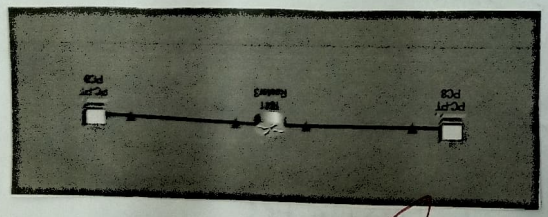
Device	IP Address	Subnet Mask
PC0	192.168.10.2	255.255.255.0
PC1	192.168.20.2	255.255.255.0

Step 3: Connecting PCs with Router.

1) connect PC0 FastEthernet 0 to Router1 FastEthernet 0/0 using a cooper straight through cable.
2) connect PC1 FastEthernet0 to Router1 FastEthernet 0/1 using a cooper straight through cable
3). Wait ~~except~~ until the interfaces becomes active.
4) Use simulation mode and send a reply PDV from PC0 to PC1
5) Verify that the PDV reaches PC1 and an acknowledgement is received back to PC0

CONFIGURATION TABLE:

Device	FastEthernet 0/0	Subnet Mask	FastEthernet 0/1	Subnet Mask
Router1.	192.168.10.1.	255.255.255.0	192.168.20.1	255.255.255.0.



RESULT:

Thus a simple Internetwork was successfully implemented on Cisco Packet TRACER

EXPERIMENT 10(b)

28/9/26 INTERNETWORK USING WIRELESS ROUTER

AIM:

To design and configure an internetwork using a wireless router, DHCP server and internet cloud in Cisco Packet Tracer.

PROCEDURE:

STEP 1:- Build the network topology.

1) Open Cisco Packet Tracer and all the devices like PC, Router, laptop, cable modem, internet cloud and server.
2). Connect the device using appropriate cables.

STEP 2:- Configure the Router.

1). Wireless Router → GUI → Wireless Network name → Homenetwork
2) Go to setup tab and ensure DHCP is enabled.
3). Configure the server as 208.67.220.220.
4) Save the settings.

STEP 3:- Configure the Laptop.

1). Laptop → Physical tab → poweroff → remove ethernet cable and add WPC300N → Power on the laptop
2). Desktop → PC wireless, select connect tab and select Homenetwork.
3) Connect the laptop to the wireless network

STEP 4:- Configure the PC.

1) PC → Desktop → IP configuration. Select DHCP
2) The PC automatically receives and IP address from the wireless router
3) Use the command prompt and ipconfig to verify the configuration

STEP 5:- Configure the Internet cloud.

1) Open the Internet cloud.
2) Config cable
3) Set :- From port: coaxial TO port: Ethernet
4) Click add.
5) Go TO Ethernet and select cable as Provider Network

STEP 6:- Configure the CISCO.com Server.

DHCP Configuration

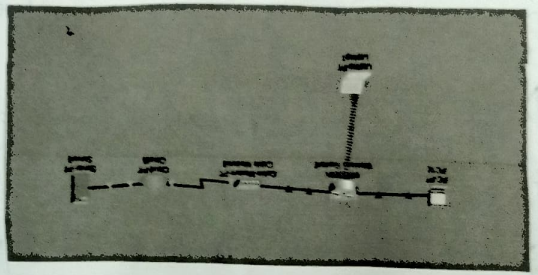
Parameter	Value
Pool Name	DHCP Pool
Default Gateway	[illegible]
DNS Server	"
Starting IP	[illegible]
Subnet Mask	[illegible]
Max User	50

DNS Configuration

Parameter	Value
Name	cisco.com
Type	A record
Address	2001:db8:2001:20

STEP 7:-

Verify Connectivity



RESULT:

Thus the wireless router, DHCP server and Internet cloud were successfully configured.